

Function Behavior & Structure

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **determine** whether a relation is a function.

RELATED SKILLS

☐ D Identify input and output values in a formula, graph, table, or context.

☐ A Apply the vertical line test to a graph.

☐ 2. I can **evaluate** a function from a formula, graph, or table.

RELATED SKILLS

☐ D Identify input and output values in a formula, graph, table, or context.

☐ A Simplify an algebraic expression using operation properties.

☐ A Solve for an input associated with a specified output.

☐ 3. I can **determine** domain and range from a formula, graph, or table.

RELATED SKILLS

☐ I Determine domain restrictions from denominators, even roots, and logarithms.

☐ D Read domain and range from a graph.

☐ D Read domain and range from a table.

☐ 4. I can **determine** algebraic domain restrictions caused by denominators and even roots.

RELATED SKILLS

☐ D Determine domain restrictions from denominators, even roots, and logarithms.

☐ D Require an even-root radicand to be nonnegative when determining domain.

☐ A Determine excluded values from an original denominator.

☐ A Factor a monic quadratic trinomial.

☐ A Factor a quadratic trinomial with a nonunit leading coefficient.

☐ 5. I can **graph** a piecewise function from its formula.

RELATED SKILLS

☐ I Graph each rule on its stated domain with correct endpoints.

☐ I Select the correct rule of a piecewise function for a given input.

☐ A Describe increasing or decreasing behavior over intervals.

☐ 6. I can **write** a piecewise formula that represents a graph.

RELATED SKILLS

☐ I Translate a segmented graph into a piecewise formula.

☐ A Convert among common forms of a linear equation.

☐ A Describe increasing or decreasing behavior over intervals.

☐ 7. I can **graph** a transformed function using shifts, stretches, compressions, and reflections.

RELATED SKILLS

☐ I Distinguish stretches from compressions and horizontal effects from vertical effects.

☐ I Interpret transformations applied to function inputs.

☐ I Interpret transformations applied to function outputs.

☐ A Connect parameters in a formula to features of its graph.

☐ 8. I can **describe** how a transformation affects domain, range, or asymptotes.

RELATED SKILLS

☐ I Exchange domain and range between a function and its inverse.

☐ D Distinguish stretches from compressions and horizontal effects from vertical effects.

☐ D Interpret transformations applied to function inputs.

☐ D Interpret transformations applied to function outputs.

☐ 9. I can **evaluate** a composition from formulas, graphs, or tables.

RELATED SKILLS

☐ I Follow intermediate values to compose functions from tables.

☐ I Substitute one function formula into another.

☐ A Identify input and output values in a formula, graph, table, or context.

☐ 10. I can **write** a formula for a composition of two functions.

RELATED SKILLS

☐ D Substitute one function formula into another.

☐ A Simplify an algebraic expression using operation properties.

☐ 11. I can **find** an inverse function algebraically.

RELATED SKILLS

- ☐ I Exchange input and output values when interpreting an inverse.
- ☐ I Exchange variables and solve to obtain an inverse formula.
- ☐ A Isolate a specified variable in an equation or formula.

☐ 12. I can **evaluate and interpret** an inverse from a graph, formula, or table.

RELATED SKILLS

- ☐ D Exchange input and output values when interpreting an inverse.
- ☐ A Exchange variables and solve to obtain an inverse formula.
- ☐ A Identify input and output values in a formula, graph, table, or context.

☐ 13. I can **determine and interpret** the units of an inverse function.

RELATED SKILLS

- ☐ I Reverse input and output units when interpreting an inverse.
- ☐ A Attach and interpret units for rates, inputs, and outputs.

☐ 14. I can **explain** how the domain and range of a function relate to those of its inverse.

RELATED SKILLS

- ☐ I Exchange domain and range between a function and its inverse.
- ☐ A Exchange input and output values when interpreting an inverse.

☐ 15. I can **determine** the range of a function by analyzing the domain of its inverse.

RELATED SKILLS

- ☐ D Exchange domain and range between a function and its inverse.
- ☐ A Determine domain restrictions from denominators, even roots, and logarithms.

☐ 16. I can **verify** algebraically that two functions are inverses.

RELATED SKILLS

- ☐ I Verify inverse functions using composition.
- ☐ A Simplify an algebraic expression using operation properties.
- ☐ A Substitute one function formula into another.

☐ 17. I can **interpret** the meaning and units of a composition.

RELATED SKILLS

- ☐ D Attach and interpret units for rates, inputs, and outputs.
- ☐ D Follow intermediate values to compose functions from tables.
- ☐ D Substitute one function formula into another.
- ☐ A Explain a model parameter in the language of its context.

☐ 18. I can **classify** a graph or table as concave up or concave down using changes in average rate of change.

RELATED SKILLS

- ☐ I Use successive average rates of change to classify concavity.
- ☐ D Use tabular patterns to predict graph shape and behavior.
- ☐ A Calculate average rate of change over an interval.

☐ 19. I can **determine** whether a function is invertible using the horizontal line test.

RELATED SKILLS

- ☐ I Apply the horizontal line test for invertibility.
- ☐ I Test whether a function is one-to-one on a domain.

☐ 20. I can **graph** an inverse function as a reflection across the line $y = x$.

RELATED SKILLS

- ☐ D Connect parameters in a formula to features of its graph.
- ☐ D Exchange input and output values when interpreting an inverse.
- ☐ A Exchange domain and range between a function and its inverse.